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Master's Thesis

**Controllable Zero-Shot Text
Generation via Soft Label
Conditioning**

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**Graduate School
Korea University**

August 2025

Controllable Zero-Shot Text Generation via Soft Label Conditioning

by

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requirements for the degree of Master of Business
Administration

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Controllable Zero-Shot Text Generation via Soft Label Conditioning

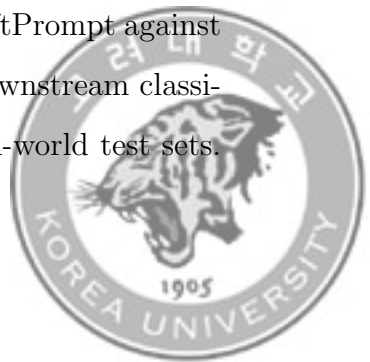
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Abstract

The scarcity of large-scale, high-quality labeled data remains a critical bottleneck in natural language processing (NLP), particularly for tasks involving nuanced or subjective human judgments. While Large Language Models (LLMs) offer a promising avenue for synthetic data generation, existing methods predominantly rely on hard-label conditioning (e.g., “positive”/“negative”), which fails to capture the continuous nature of many linguistic phenomena and often yields polarized or homogenous datasets. To address these limitations, we introduce SoftPrompt, a novel framework that conditions LLMs on soft labels—continuous values sampled from distributions (e.g., Beta, Dirichlet)—to generate synthetic text. By enabling prompts like “Generate a review that is 70% positive,” SoftPrompt facilitates a more expressive and realistic coverage of the label space, directly producing (text, soft label) pairs for downstream soft supervision. We evaluate SoftPrompt against a hard-label conditioned baseline, assessing its impact on downstream classification performance and textual diversity using several real-world test sets.



Our results demonstrate that SoftPrompt significantly outperforms hard-label generation in terms of generalization accuracy and data diversity, especially in data-limited or subjective settings. Furthermore, we find that strategically excluding the most ambiguous soft-labeled examples (those near borderline) can further enhance downstream model robustness. SoftPrompt offers a more nuanced approach to synthetic data generation, paving the way for more effective training of models on complex NLP tasks.

Keywords: Large language models, controllable text generation, zero-shot learning, soft labels, synthetic data generation, prompt engineering



본 논문을 작성하는 데 있어 많은 분들의 도움을 받았습니다. 이 글을 통해 그동안 전하지 못한 감사의 마음을 전하고자 합니다.

먼저, 항상 큰 힘이 되어준 사랑하는 가족에게 깊은 감사와 사랑의 마음을 전합니다.

다음으로, 부족한 저를 세심하게 지도해주시고 아낌없는 격려와 조언을 통해 연구자로 성장할 수 있도록 이끌어주신 강현구 지도교수님께 진심으로 감사드립니다. 교수님의 첫 제자가 될 수 있어 더없이 영광이었으며, 교수님과 함께한 시간은 단순한 연구 과정이 아니라 제 인생에 있어 뜻깊고 소중한 성장의 기회였습니다. 언제나 따뜻한 배려와 격려로 용기와 자신감을 심어주셨기에 오늘의 제가 있을 수 있었습니다. 교수님을 본받아 더욱 성숙하고 책임감 있는 연구자가 될 수 있도록 끊임없이 노력하겠습니다.

어려울 때마다 위로와 격려를 아끼지 않으며 큰 힘이 되어준 왕약택 석사에게도 깊은 감사를 드립니다. 같은 연구자로서뿐 아니라 소중한 친구로서 함께한 모든 추억을 잊지 않겠습니다.

항상 저를 응원해준 세라, 채원, 선규에게도 진심으로 고맙다는 말을 전하고 싶습니다.

마지막으로, 힘든 순간을 보낼 때 마다 포기하지 않고 끝까지 나아갈 수 있도록 도와주신 오세현 선생님께 깊은 감사를 드립니다. 석사 과정뿐만 아니라 제 삶 전체에 걸쳐 커다란 힘이 되어주셨습니다. 늘 저의 행복과 안정을 바라주셨던 만큼, 선생님께서도 항상 평안하고 행복하시길 진심으로 바랍니다.

모든 분께 다시 한번 진심으로 감사드립니다.



Preface

It is widely considered that the most common method for improving the performance of machine learning models is to increase the amount of training data. However, this seemingly simple solution confronts a significant practical constraint: the immense time and cost required to collect high-quality data.

When I began this research, interest in Large Language Models (LLMs) was surging. I saw the potential to overcome the persistent challenge of data collection and effectively augment training data by leveraging these powerful tools to generate realistic datasets. My initial attempts were simple, involving commands to generate specific types of sentences. However, this soon led me to a more fundamental question: "How can we make machine-generated data as natural as human-created data?"

I sought to imbue the data with the subtlety and uncertainty of real-world language—its ambiguous, non-discrete boundaries. It was at this juncture that the concept of ‘Soft Labeling,’ and our research ‘SoftPrompt’, was born. We proposed a new paradigm: conditioning the model on flexible probabilistic values, such as “a review that is 70% positive”, instead of definitive hard labels. This was an attempt to overcome the limitations of existing data generation methods based on hard labels. The data generated through this idea better reflected the diversity and nuance of natural language, and in turn, downstream models trained on it demonstrated superior generalization performance and stability.

The research process, of course, was not without its challenges. Attempts



to more precisely control certain attributes, such as diversity, sometimes yielded unexpected results. Through this process, however, we came to appreciate the complexity and depth of controlling large language models, and I am confident that these experiences have become a solid foundation for future research.

This paper would not have been possible without the help of many individuals. I would first like to express my deepest gratitude to Professor Hyungu Kahng, who guided the direction of this research with his profound insights and warm encouragement.

I hope this paper offers a small inspiration to the many researchers grappling with similar challenges, and that it may contribute, even in a small way, to the advancement of artificial intelligence technology that can more deeply understand our world.



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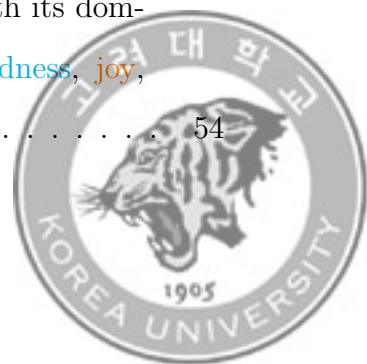


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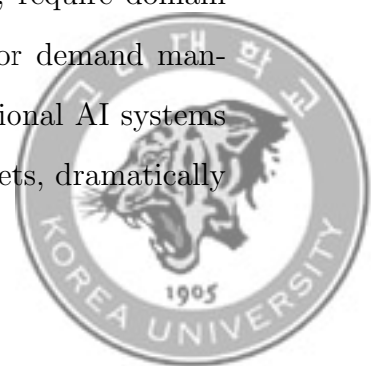
Chapter 1

Introduction

1.1 Background and Motivation

The success of modern supervised learning paradigms depends on access to large-scale, high-quality labeled datasets. However, the acquisition of such data remains a critical bottleneck, constrained by high costs, time, privacy concerns, and domain-specific difficulties [1, 2]. While synthetic data generation using Large Language Models (LLMs) has emerged as a promising solution to this scarcity [3, 4], its purpose extends far beyond a passive remedy. We argue that synthetic data generation is an active and powerful technology for proactively designing AI capabilities and creating new business value.

This approach can accelerate the development of solutions for specific business problems where data collection is challenging. For example, labeled corpora may be scarce for tasks in low-resource languages [5], require domain expertise in sensitive fields like legal or medical NLP [6], or demand manual curation of emotionally diverse utterances for conversational AI systems [7]. Synthetic generation can instantly create tailored datasets, dramatically



shortening development cycles. This is particularly relevant for capturing the nuanced and subjective nature of human language, which is often underrepresented in existing datasets [8, 9].

Furthermore, high-quality synthetic data serves as an essential ingredient for fine-tuning and instruction-tuning LLMs. The performance of modern LLMs is increasingly defined not by their stored knowledge, but by their ability to follow instructions with precision. This requires vast datasets of high-quality (instruction, output) pairs, a demand that synthetic data is uniquely positioned to meet. Consider developing an AI writing assistant for journalists that must control for subtle tones. Using our SoftPrompt framework, one could generate thousands of sophisticated instruction pairs, such as:

- **Instruction:** *"Rewrite this factual news draft with a tone that is 80% objective but 20% cautiously optimistic."*
- **Output:** *A rewritten article reflecting this nuanced requirement.*

By fine-tuning an LLM on such a dataset, one could create a highly specialized tool capable of nuanced tonal control far beyond the binary distinctions of existing systems. This unlocks a new level of model controllability. Thus, this thesis is motivated by the need to move beyond generating data for quantity, and instead asks: how can we generate data that is qualitatively realistic, diverse, and, most importantly, precisely controlled to meet specific, nuanced objectives?



1.2 Research Problem and Objectives

Despite the promise of LLM-generated data, current methods predominantly rely on hard-label prompting—conditioning the model on a discrete class label (e.g., “positive” or “negative”). While effective in some contexts, this approach imposes overly rigid boundaries on tasks that naturally exhibit gradation and ambiguity, such as sentiment, emotion, or stance classification [10, 11]. Hard labels limit the diversity of generated text and often yield synthetic datasets that are either unrealistically polarized or homogenous, reducing their utility in downstream model training [12].

In this thesis, we aim to address these limitations by introducing a novel synthetic data generation method based on **soft label conditioning**. Rather than assigning binary or categorical labels to text, our method—named **SoftPrompt**—uses real-valued label vectors sampled from continuous probability distributions (e.g., Beta, Dirichlet). These soft labels are then encoded into natural language prompts (e.g., “Generate a review that is 70% positive”), guiding the LLM to generate text that reflects more nuanced or mixed characteristics.

1.3 Contributions

This thesis proposes a new framework for LLM-based data generation that improves upon hard-label prompting in several key ways:

- We propose **SoftPrompt**, a framework that conditions large language models on soft labels sampled from continuous distributions.

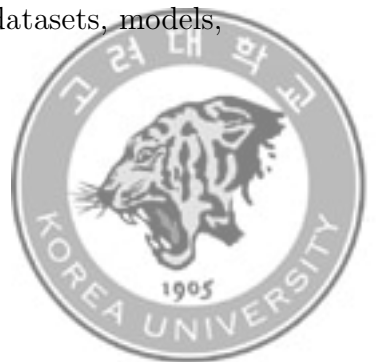


- We show how to convert these label vectors into meaningful natural language prompts to control generation behavior.
- We demonstrate, through experiments on multiple NLP datasets, that SoftPrompt yields better generalization accuracy and diversity than hard-label generation.
- We introduce margin-based filtering to exclude highly ambiguous (e.g., near 0.5) soft labels and show that this improves downstream classification robustness.
- We evaluate the effect of training models directly on soft labels, providing evidence that this leads to better calibration.

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

- Chapter 2 provides background on synthetic data generation and reviews related work on hard/soft labeling and LLM prompting techniques.
- Chapter 3 introduces the proposed SoftPrompt framework, including its design motivation, label sampling strategy, and prompt formulation.
- Chapter 4 outlines the experimental setup, including datasets, models, and evaluation metrics.



- Chapter 5 presents and analyzes experimental results on classification accuracy, robustness, and text diversity.
- Chapter 6 discusses key findings, limitations of the current work, and directions for future research.
- Chapter 7 concludes the thesis with a summary of contributions.



Chapter 2

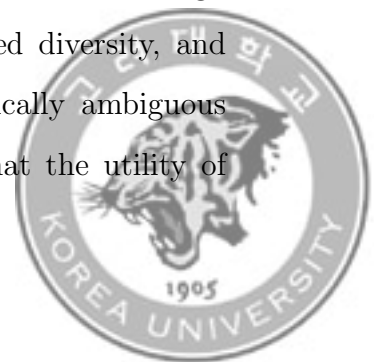
Related Work

2.1 Synthetic data generation with LLMs

Large language models (LLMs) have unlocked new opportunities for synthetic data generation, offering scalable alternatives to costly manual annotation [6]. However, prevailing approaches condition LLMs on hard labels—fixed discrete categories such as sentiment polarity or topic class—to guide text generation [3, 9]. While this strategy ensures task alignment, it imposes rigid boundaries on inherently continuous linguistic phenomena, limiting semantic richness, diversity, and generalization capacity [12].

2.2 Limitations of hard-label conditioning

Empirical studies have shown that synthetic datasets created through hard-label prompting suffer from systematic biases, reduced diversity, and poor generalization, particularly on subjective or semantically ambiguous tasks [3, 12]. For example, Li et al. [9] demonstrated that the utility of



synthetic data diminishes significantly as task subjectivity increases—a finding echoed across multi-class and fine-grained classification settings. These results collectively highlight the inadequacy of hard-label conditioning for faithfully modeling real-world linguistic complexity.

2.3 Soft labels: From Mixup to prompt conditioning

The idea of soft labels—continuous rather than categorical supervision—has roots in the Mixup method from computer vision [13], which constructs virtual samples by linearly interpolating both inputs and labels. This yields soft targets that promote generalization and robustness.

GPT3Mix [14] adapts this idea to NLP by generating synthetic text using large-scale LLMs and assigning soft labels post-hoc, derived from model-predicted label token distributions. A critical aspect of GPT3Mix is that it is inherently a few-shot approach, relying on providing in-context examples to the LLM to demonstrate how to interpolate between texts. This dependency introduces practical challenges, including additional data curation costs and, as we discuss in Chapter 5, a potential reduction in output diversity due to the LLM’s tendency to imitate the provided examples.

In contrast, our method, SoftPrompt, operationalizes soft labels as a pre-hoc, active control signal. By directly embedding the soft label into the prompt before generation, our framework marks a fundamental departure from both Mixup and GPT3Mix. Rather than interpolating existing data to



infer soft labels, we interpolate abstract labels to generate entirely new data in a zero-shot manner. This design enables LLMs to model semantic ambiguity and human-like subjectivity more naturally, without the constraints of imitation.

2.4 Prompt engineering and reasoning for high-quality generation

Beyond label strategies, prompt design plays a pivotal role in determining the quality and reliability of synthetic data. Structured prompting techniques, such as zero-shot chain-of-thought (CoT) reasoning, have been shown to better activate latent model capabilities compared to naive prompting [15, 16]. Moreover, reasoning-augmented LLMs generate synthetic datasets with greater semantic fidelity and distributional robustness than non-reasoning counterparts, underscoring the importance of combining expressive supervision with prompt structures that foster deeper reasoning [10].

AttrPrompt [3], for example, enhances diversity and mitigates bias by conditioning generation on various auxiliary attributes (e.g., sentence length, tone, locality). While this improves surface-level variation, such approaches are still grounded in discrete conditioning. In contrast, SoftPrompt introduces continuous label supervision as a semantically grounded control signal, which serves both diversity and alignment objectives simultaneously.



2.5 The need for fine-grained control in synthetic data generation

Promoting diversity without sacrificing label fidelity remains a key challenge in synthetic data generation. Techniques such as temperature scaling and logit suppression have been used to diversify outputs [17], but often degrade label accuracy, necessitating costly human interventions like label replacement. Recent best practices emphasize that semantically meaningful control signals—not random noise injection—are essential for producing synthetic datasets that maintain both diversity and task alignment [3, 4, 4]. SoftPrompt contributes to this effort by introducing label-continuous, probabilistic guidance during generation, supporting fine-grained supervision in low-resource or subjective tasks.



Chapter 3

Proposed Framework: SoftPrompt

3.1 Overview and Motivation

The scarcity and expense of real labeled data frequently limit progress in supervised learning tasks, particularly in domains that require domain expertise or involve sensitive data [10]. Generative approaches have emerged as a potential remedy by synthesizing labeled data that can augment or substitute for real data [3]. To this end, we propose SoftPrompt, a novel *zero-shot* framework which produces synthetic data endowed with *soft-labeled* supervision. A central feature of our approach is the ability to specify and control the label distribution in a flexible, continuous manner, rather than forcing each sample to be deterministically drawn from a single class. This design contrasts with simpler methods that place all probability mass on one discrete label for each generated text.



3.2 Problem Setup

Consider a classification task with a label set $\mathcal{Y} = \{1, \dots, K\}$, where $K = 2$ for binary classification or $K > 2$ for multiclass classification. Our objective is to generate a synthetic dataset $\tilde{\mathcal{D}}$ of size N , consisting of pairs $(\mathbf{x}_i, y_i)$. In each pair, $\mathbf{x}_i$ represents a text string sampled from a large language model (LLM), and y_i encodes the corresponding label information.

Formally, we assume that each synthetic example $(\mathbf{x}_i, y_i)$ is generated via a three-stage process: (i) *label specification*, (ii) *prompt construction*, and (iii) *text generation*.

First, in the label specification stage, a label variable y_i is sampled from a distribution $\mathcal{P}(y)$. This distribution determines the nature of the label. Depending on the specific generation method used (detailed in subsequent sections), this step may yield either a single discrete label $y_i \in \mathcal{Y}$ or a probability vector $y_i \in \Delta^{K-1}$, which is the $(K - 1)$ -probability simplex.

Second, the prompt construction stage maps the sampled label specification y_i to a natural language instruction $\pi_i \in \mathcal{T}_{\text{prompt}}$. This is achieved using a predefined mapping $\pi(\cdot)$, such that $\pi_i = \pi(y_i)$. The prompt π_i is designed to guide the LLM towards generating text consistent with the specified label information y_i .

Third, in the text generation stage, the constructed prompt π_i is input to the LLM, denoted by $G : \mathcal{T}_{\text{prompt}} \rightarrow \mathcal{P}(\mathcal{T})$. The LLM G defines a probability distribution $G(\cdot | \pi_i)$ over the space of possible textual outputs $\mathcal{T}$, conditioned on the prompt. A text string $\mathbf{x}_i$ is then sampled from this distribution: $\mathbf{x}_i \sim G(\cdot | \pi_i)$.



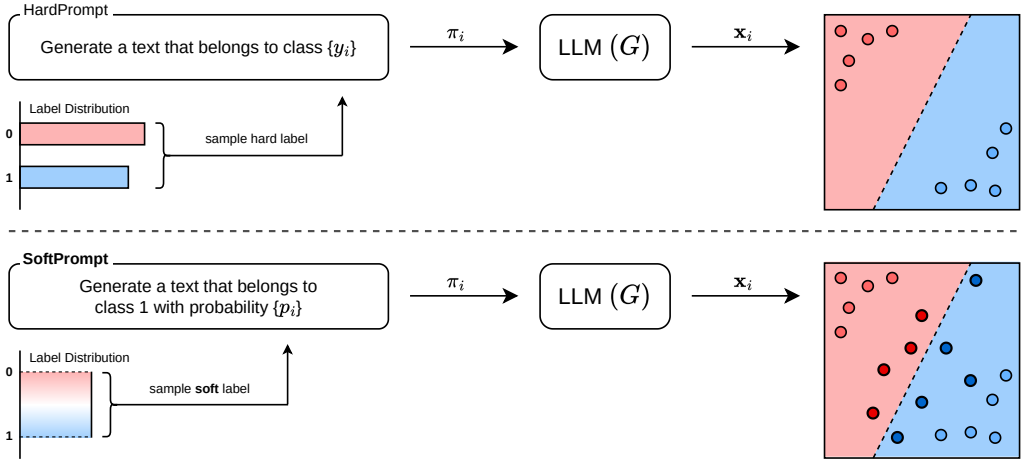


Figure 3.1: Illustration of the HardPrompt baseline versus our proposed SoftPrompt framework, for binary tasks. (Top) HardPrompt utilizes hard labels (e.g., y_i) to generate text. (Bottom) Our proposed **SoftPrompt** framework employs continuous soft labels (e.g., probability p_i) sampled from a distribution to guide the LLM (G), enabling more nuanced synthetic text ($\mathbf{x}_i$) generation.

Consequently, the resulting synthetic data pair is $(\mathbf{x}_i, y_i)$, where the exact form of y_i (discrete scalar or probability vector) depends on the specifics of the label specification stage. Repeating this three-stage process N times yields the synthetic dataset $\tilde{\mathcal{D}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, which can then be used to train a downstream classifier. In what follows, we first present a baseline approach based on discrete label specification, and then introduce our proposed method utilizing continuous label specifications.



3.3 Hard-Label Baseline

Following prior work in LLM-based text generation [3, 10, 12], we establish a baseline approach, termed **HardPrompt**, which conditions the LLM on a single discrete class label $y_i \in \mathcal{Y}$. The process begins with *label specification*: for each target synthetic example i , a label y_i is sampled from either a Bernoulli(θ) distribution (for binary classification, $K = 2$) or a Categorical($\boldsymbol{\theta}$) distribution (for multiclass, $K > 2$). The parameters $\theta \in [0, 1]$ or $\boldsymbol{\theta} \in \Delta^{K-1}$ are chosen beforehand to control the overall class frequencies in the generated dataset.

Subsequently, this discrete label y_i is used for *prompt construction*, defining a textual instruction $\pi(y_i)$ that explicitly directs the LLM to generate text belonging only to that specific class. For instance, in a binary sentiment classification task, if y_i represents the ‘positive’ class, the prompt might be $\pi(y_i) = \text{“Generate a positive review”}$. Finally, this hard-label prompt $\pi(y_i)$ is provided to the LLM G , from which the text realization $\mathbf{x}_i$ is sampled.

Repeating this procedure N times yields a hard-labeled synthetic dataset $\tilde{\mathcal{D}}_{\text{hard}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where each y_i is a single class identifier (e.g., 0 or 1 in the binary case). Although straightforward to implement, HardPrompt is inherently limited: it cannot represent mixed or continuous attributes (e.g., “70% positive”) and may thus fail to explore potentially diverse regions of the true label space, forcing generated examples into strict, potentially unrealistic categorical bins.



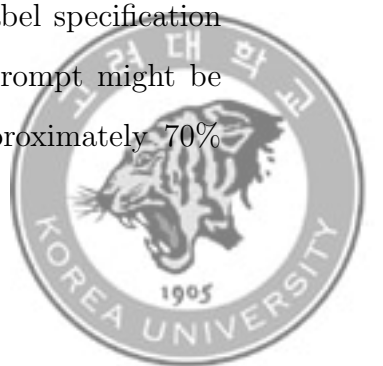
3.4 Proposed Method: SoftPrompt

We propose **SoftPrompt**, a framework designed to overcome the limitations inherent in assigning a single discrete label to each generated text, as done in HardPrompt. The core innovation of our approach is the generation of synthetic samples endowed with *soft-labeled supervision*, enabling the representation of partial class memberships or attribute intensities. Figure 3.1 illustrates the overall process of our proposed framework.

Continuous label specification. Instead of sampling a discrete label $y_i \in \mathcal{Y}$, SoftPrompt begins by sampling a probability vector $\mathbf{p}_i \in [0, 1]^K$ such that $\sum_{k=1}^K p_{ik} = 1$. For binary classification ($K = 2$), this vector is $\mathbf{p}_i = (1 - p_i, p_i)$, where the scalar probability p_i can be drawn from a flexible distribution, such as $\text{Beta}(\alpha, \beta)$ or another user-specified distribution over $[0, 1]$. In the multiclass setting ($K > 2$), $\mathbf{p}_i$ can similarly be drawn from a suitable distribution over the simplex, like $\text{Dirichlet}(\boldsymbol{\alpha})$, or via alternative schemes tailored to the specific task. Formally, the label specification step involves sampling:

$$\mathbf{p}_i \sim \mathcal{P}(\mathbf{p}), \quad \text{where } \mathcal{P}(\mathbf{p}) \text{ is a distribution supported on } \Delta^{K-1}.$$

Prompt construction. The core of SoftPrompt lies in constructing a natural language prompt $\pi(\mathbf{p}_i)$ that encodes the continuous label specification represented by $\mathbf{p}_i$. For binary sentiment classification, a prompt might be constructed as: “Generate a movie review expressing approximately 70%



positivity and 30% negativity.” For a multiclass task like the 6-way Emotion dataset (sadness, joy, love, anger, fear, surprise), given a sampled vector (e.g., emphasizing ’joy’ primarily), the prompt could be formulated as: “Generate a sentence expressing primarily Joy (around 50%), with slight undertones of other emotions (around 10% each: sadness, love, anger, fear, surprise).” While perfect adherence to these numerical specifications by the LLM cannot be guaranteed, this style of prompting guides the LLM towards generating text that reflects the intended distribution of attributes.

Text generation. Given the soft label vector $\mathbf{p}_i$ and its corresponding prompt $\pi(\mathbf{p}_i)$, the text $\mathbf{x}_i$ is generated by sampling from the LLM:

$$\mathbf{x}_i \sim G(\cdot | \pi(\mathbf{p}_i)).$$

The resulting pair $(\mathbf{x}_i, \mathbf{p}_i)$ constitutes one sample in the synthetic dataset. Repeating this N times yields the complete soft-labeled dataset: $\tilde{\mathcal{D}}_{\text{soft}} = \{(\mathbf{x}_i, \mathbf{p}_i)\}_{i=1}^N$. For downstream model training, the soft label vector $\mathbf{p}_i$ can be discretized into a one-hot label based on the highest probability component or, if necessary, used directly as a target distribution (e.g., minimizing KL divergence against soft labels). We explore the benefits of training with soft targets in Section 4.4.

SoftPrompt provides richer supervision by explicitly modeling the potential for borderline cases or mixed sentiments often found in real-world data (e.g., a statement that is simultaneously somewhat positive and somewhat negative). Furthermore, this design enables fine-grained control over attributes, allowing



for systematic exploration of attribute intensities or targeted sampling across the label simplex. Sampling from appropriately chosen distributions $\mathcal{P}(\mathbf{p})$ facilitates emphasizing underrepresented attribute levels or generating examples of rare mixtures potentially absent in available real data. SoftPrompt thus addresses both the lack of nuanced coverage in HardPrompt and the need for flexible data augmentation strategies.



Chapter 4

Experiments, Results, and Analysis

This section presents a comprehensive empirical evaluation of our proposed SoftPrompt framework. We assess its efficacy against several baselines—including HardPrompt and Gold standard (training on real data)—across a diverse set of binary and multiclass text classification tasks. Our experiments adhere to a consistent evaluation protocol (detailed in Section 4.1) and utilize five standard benchmark datasets: IMDb [18], SST [11], and SUBJ [8] for binary classification, and Emotion [19] and AGNews [20] for multiclass classification.

Our analysis focuses on several key aspects: (i) the comparative performance of SoftPrompt against HardPrompt and Gold models; (ii) the impact of varying synthetic training data sizes; (iii) the effectiveness of a margin-based filtering technique to refine synthetic datasets by excluding examples derived from ambiguous soft label scores; and (iv) the benefits of directly training downstream models with soft label targets to improve performance



and model calibration. All reported performance metrics are averaged over 50 bootstrap trials to ensure robust estimations.

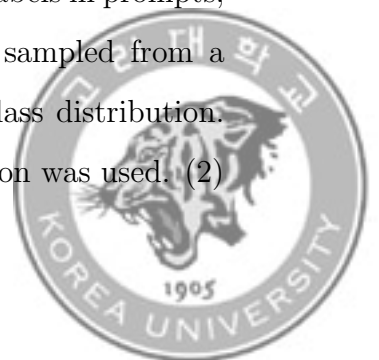
4.1 Experimental setup

Table 4.1: Summary of datasets used in our experiments. Sizes refer to the original benchmark splits.

Characteristic	IMDb	SST	SUBJ	Emotion	AGNews
Task Type	Sentiment	Sentiment	Subjectivity	Emotion	Topic
K (No. classes)	2	2	2	6	4
N_{train}	25000	6920	8000	89832	120000
N_{test}	25000	1821	2000	20000	7600

Datasets. Table 4.1 provides an overview of the benchmark datasets used. For each dataset, we utilize its standard test split for evaluation. To investigate the impact of synthetic data quantity, we generate synthetic training datasets of varying sizes, specifically $r \times N_{\text{train}}$, where N_{train} is the size of the original real training set for the respective benchmark, and $r \in \{1\%, 5\%, 10\%, 25\%, 50\%, 100\%\}$.

Baselines. We compare SoftPrompt against two primary baselines: (1) **HardPrompt** generates synthetic data using discrete class labels in prompts, as detailed in Section 3.3. For binary tasks, labels were sampled from a Bernoulli distribution with $\theta = 0.5$ to ensure a balanced class distribution. For multiclass tasks, a similar uniform categorical distribution was used. (2)



Gold utilizes the corresponding proportion of the actual human-labeled training data from the benchmark dataset. This serves as an approximate upper bound for performance with data of equivalent size.

Label specification for SoftPrompt. For binary classification tasks using SoftPrompt, the continuous label p_i (representing the probability of the positive class) was sampled from a Uniform distribution, $p_i \sim U(0, 1)$. For multiclass tasks, the soft label vector $\mathbf{p}_i$ was sampled from a Dirichlet distribution, $\mathbf{p}_i \sim \text{Dirichlet}(\boldsymbol{\alpha} = \mathbf{1})$.

LLM, prompt design, and CoT strategy. All synthetic data, for both hard and soft conditions, is generated using `gemini-2.0-flash`. This model was selected for its balance of strong performance and affordability. We fix the temperature to 1.0. Crucially, to isolate the effect of soft versus hard labeling, we intentionally kept the prompts for both methods as simple as possible, providing minimal instructions beyond the label specification itself (see Appendix ??). This was done to mitigate biases that overly specific or engineered prompt instructions might introduce. Furthermore, to assess whether eliciting a reasoning step prior to text generation enhances the quality of synthetic data, we conduct experiments where a Chain-of-Thought (CoT) instruction (e.g., “Think step-by-step before generating the text.”) is prepended to these simple prompts [21].

Feature representation and downstream task setup. For all datasets, text instances are converted into fixed-size vector representations using the OpenAI’s `text-embedding-3-small` model via its API. These 1536-dimensional



embeddings serve as input features for the downstream classifier. To ensure a controlled comparison and primarily isolate the effects of data quality rather than complex model capacity, we employ a logistic regression model as our primary downstream classifier. For initial experiments comparing SoftPrompt with HardPrompt (in Section 4.2), the soft labels generated by SoftPrompt (i.e., probability vectors p_i) are converted to hard labels by taking the argmax over the probability vector components. This allows for a direct comparison using standard classification metrics and training procedures.

Metrics. Performance is primarily measured by accuracy for the IMDB, SST, SUBJ, and AGNews datasets. For the Emotion dataset, which is a multiclass task with class imbalances in its test set, we use the macro F1-score as the primary evaluation metric. As noted, all key performance metrics reported are averages over 50 bootstrap trials of training the downstream classifier on different (sub)samples of the generated or real data, ensuring the stability of our conclusions.



Table 4.2: Downstream classification performance under the evaluation protocol described in Section 4.1, SoftPrompt and baselines at various synthetic data sizes. “Gold” indicates training on real data. The best value in each configuration is **boldfaced**, and the second best is underlined.

Method	Size (%)	IMDb	SST	SUBJ	Emotion	AGNews
Gold		91.5 (± 0.7)	83.5 (± 2.1)	92.5 (± 1.1)	58.4 (± 0.6)	89.0 (± 0.5)
HardPrompt		73.6 (± 9.9)	69.6 (± 9.8)	55.1 (± 5.0)	26.6 (± 0.5)	58.8 (± 1.4)
HardPrompt+CoT	1%	<u>82.9</u> (± 1.3)	80.7 (± 1.7)	69.8 (± 1.9)	33.5 (± 0.7)	71.8 (± 0.5)
SoftPrompt		75.2 (± 4.7)	69.1 (± 6.1)	<u>71.2</u> (± 8.0)	38.3 (± 2.6)	57.7 (± 3.4)
SoftPrompt+CoT		86.6 (± 2.3)	<u>75.5</u> (± 3.2)	71.5 (± 7.0)	<u>38.2</u> (± 2.1)	<u>63.2</u> (± 5.0)
Gold		93.1 (± 0.3)	87.3 (± 0.9)	94.7 (± 0.5)	66.2 (± 0.3)	90.7 (± 0.1)
HardPrompt		78.9 (± 8.8)	69.4 (± 11.8)	56.0 (± 5.5)	26.9 (± 0.4)	57.1 (± 0.8)
HardPrompt+CoT	5%	<u>82.8</u> (± 0.7)	81.3 (± 0.5)	70.2 (± 1.1)	33.9 (± 0.5)	72.0 (± 0.1)
SoftPrompt		76.1 (± 4.7)	70.3 (± 4.1)	<u>73.2</u> (± 6.6)	38.7 (± 1.9)	62.2 (± 4.0)
SoftPrompt+CoT		86.8 (± 1.9)	<u>75.3</u> (± 2.6)	76.0 (± 5.0)	<u>35.8</u> (± 2.0)	<u>63.3</u> (± 3.4)
Gold		93.4 (± 0.2)	88.0 (± 0.5)	95.2 (± 0.3)	69.4 (± 0.2)	91.0 (± 0.1)
HardPrompt		79.9 (± 7.5)	72.6 (± 12.8)	58.0 (± 3.0)	26.5 (± 0.2)	56.7 (± 0.7)
HardPrompt+CoT	10%	<u>82.9</u> (± 0.7)	81.4 (± 0.3)	70.2 (± 0.9)	34.1 (± 0.5)	72.2 (± 0.3)
SoftPrompt		76.2 (± 3.9)	71.1 (± 4.1)	<u>72.5</u> (± 5.1)	39.1 (± 2.6)	62.2 (± 2.6)
SoftPrompt+CoT		87.9 (± 1.5)	<u>75.5</u> (± 2.9)	74.4 (± 5.1)	<u>36.8</u> (± 2.8)	<u>66.7</u> (± 2.1)
Gold		93.7 (± 0.3)	88.2 (± 0.7)	95.7 (± 0.3)	71.7 (± 0.2)	91.5 (± 0.1)
HardPrompt		79.9 (± 5.7)	74.1 (± 11.6)	59.0 (± 4.3)	26.8 (± 0.2)	54.7 (± 1.1)
HardPrompt+CoT	25%	<u>82.6</u> (± 0.4)	81.5 (± 0.1)	<u>70.7</u> (± 0.9)	34.0 (± 0.5)	72.1 (± 0.1)
SoftPrompt		77.2 (± 2.9)	72.4 (± 3.9)	70.1 (± 6.1)	42.1 (± 1.2)	61.8 (± 2.5)
SoftPrompt+CoT		88.9 (± 0.7)	<u>76.5</u> (± 2.7)	74.0 (± 5.2)	<u>39.1</u> (± 0.9)	<u>66.1</u> (± 1.2)
Gold		93.8 (± 0.2)	87.5 (± 1.1)	95.9 (± 0.3)	73.4 (± 0.2)	91.6 (± 0.2)
HardPrompt		80.4 (± 7.7)	74.0 (± 11.9)	59.4 (± 3.7)	26.3 (± 0.9)	57.2 (± 0.2)
HardPrompt+CoT	50%	<u>82.6</u> (± 0.5)	81.5 (± 0.1)	<u>70.1</u> (± 0.5)	34.0 (± 0.7)	72.1 (± 0.1)
SoftPrompt		77.1 (± 3.0)	73.8 (± 3.1)	69.7 (± 4.3)	41.3 (± 1.0)	61.0 (± 2.2)
SoftPrompt+CoT		89.1 (± 0.7)	<u>76.7</u> (± 2.0)	72.0 (± 6.3)	<u>37.9</u> (± 0.7)	<u>64.0</u> (± 3.9)
Gold		94.1 (± 0.3)	87.3 (± 1.0)	96.2 (± 0.3)	74.9 (± 0.3)	91.7 (± 0.1)
HardPrompt		81.8 (± 6.0)	77.4 (± 8.9)	59.5 (± 1.7)	26.7 (± 0.1)	57.3 (± 0.4)
HardPrompt+CoT	100%	<u>82.5</u> (± 1.0)	81.5 (± 0.0)	<u>69.4</u> (± 0.6)	34.0 (± 0.2)	72.0 (± 0.1)
SoftPrompt		76.6 (± 1.9)	75.2 (± 2.5)	67.9 (± 4.2)	41.2 (± 0.4)	62.4 (± 0.3)
SoftPrompt+CoT		89.3 (± 0.3)	<u>77.5</u> (± 1.7)	69.9 (± 5.9)	<u>40.2</u> (± 4.6)	<u>67.1</u> (± 0.3)

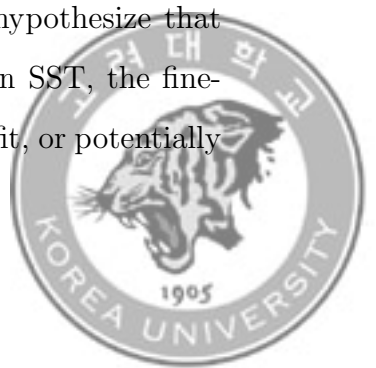


4.2 Initial performance analysis

We first evaluated models trained on synthetic data generated by SoftPrompt (with binarized labels) and HardPrompt, comparing them against the Gold standard, across different data proportions (r). The results are presented in Table 4.2.

IMDb. SoftPrompt+CoT consistently outperformed its hard-labeled counterpart across all tested data proportions (r). It also approached the performance of the Gold baseline, particularly at larger r values. For instance, at $r = 100\%$ (representing a synthetic dataset size equivalent to the full real training set), SoftPrompt+CoT achieved an accuracy of 89.3, compared to 82.5 for HardPrompt+CoT and 94.1 for the Gold baseline. Enabling CoT prompting, which may encourage the LLM to generate more structured or coherent text better aligned with the prompt’s intent, generally provided further improvements for both base versions.

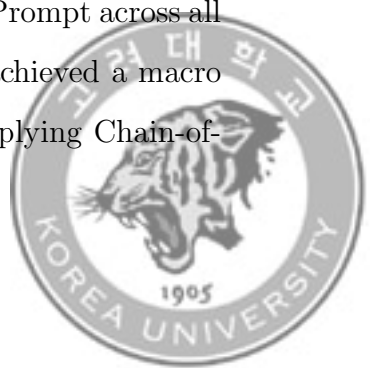
SST. Primarily consisting of shorter sentences compared to IMDb, SoftPrompt+CoT did not consistently outperform HardPrompt+CoT. At $r = 100\%$, SoftPrompt+CoT achieved an accuracy of 77.5, while HardPrompt+CoT reached 81.5. Both methods lagged behind the Gold baseline (87.3). The application of CoT prompting did not yield substantial or consistent gains for either prompting methods on this particular dataset. We hypothesize that for shorter, more direct expressions of sentiment as found in SST, the fine-grained control offered by SoftPrompt may provide less benefit, or potentially



introduce noise if the LLM struggles to map soft scores to concise text effectively. Hard labels might be a more robust guide for generating shorter, clearly polarized snippets.

SUBJ. The base SoftPrompt method demonstrated notably better performance compared to the base HardPrompt method. For instance, at a data proportion of $r = 1\%$, SoftPrompt achieved an accuracy of 71.2, outperforming HardPrompt by 16.1 percentage points. This trend of SoftPrompt surpassing HardPrompt was generally consistent across the different data proportions evaluated for their base versions. Both approaches also benefited from CoT prompting; HardPrompt+CoT saw a considerable improvement over HardPrompt, and SoftPrompt+CoT showed a slight gain over SoftPrompt. However, a critical observation for this dataset was that as the amount of soft-labeled training data increased, the predictive performance eventually began to decline. This counterintuitive trend led us to hypothesize that synthetic examples generated with soft labels near the decision boundary (i.e., ambiguous subjectivity scores) might be of lower quality or act as noise when binarized, thereby negatively impacting the classifier’s generalization. This phenomenon is explored further in our analysis of margin-based filtering (Section 4.3).

Emotion. For the Emotion dataset, which involves classifying text into one of six emotions, SoftPrompt consistently outperformed HardPrompt across all data proportions. For example, at $r = 100\%$, SoftPrompt achieved a macro F1 of 41.2 compared to 26.7 for HardPrompt. However, applying Chain-of-



Thought (CoT) prompting did not provide further significant improvements for SoftPrompt; SoftPrompt+CoT (40.2) performed similarly at $r = 100\%$. We conjecture that generating text to express mixed emotions via soft labels might rely more on direct lexical and stylistic choices rather than complex, multi-step reasoning that CoT typically encourages. For such tasks, the “step-by-step” thinking might not add substantial value or could even lead the LLM to produce less natural or less clearly emotional text.

AGNews. In contrast to other datasets, SoftPrompt’s performance on AGNews (classifying news headlines into topics) was notably lower than HardPrompt’s and the Gold baseline, even when CoT was applied. For example, at $r = 100\%$, SoftPrompt+CoT achieved 67.1 accuracy while HardPrompt+CoT reached 72.0. We hypothesize this might be related to the nature of the AGNews task and the intuition behind applying soft labels. Unlike sentiment, subjectivity, or emotion, which can be conceptualized on a continuous spectrum, news headlines are typically focused on a single discrete topic. Attempting to generate headlines with mixed topic probabilities (e.g., 30% World, 70% Sports) might be less natural for the LLM to execute coherently, especially for concise texts like headlines. The very notion of a soft-labeled news topic might be less aligned with how such data is typically generated or perceived, potentially leading to less representative synthetic examples when forced into a single topic via argmax for training. This suggests that the applicability or the precise formulation of soft prompts may need to be adapted for tasks with inherently discrete and less blendable label semantics.



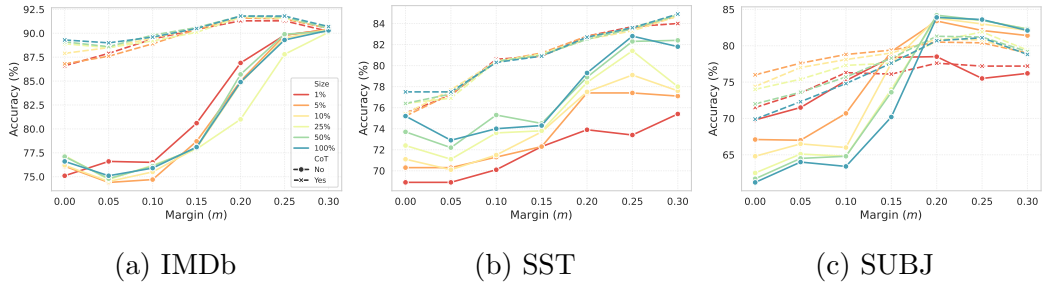


Figure 4.1: Impact of margin-based filtering on downstream classification accuracy on binary tasks. Both with and without CoT, model performance generally increases with larger margins up to an optimal range (0.20-0.25), beyond which improvements diminish or performance can slightly decline.

4.3 Improving performance with margin-based filtering

Our initial findings, particularly the performance degradation on SUBJ with increasing amounts of unfiltered synthetic data (Section 4.2), highlighted that not all SoftPrompt-generated examples offer equal utility when their labels are binarized for downstream training. This observation motivated an investigation into a margin-based filtering strategy. Rather than viewing this as a correction for a flawed generation process, we consider it an insight into optimizing the use of nuanced synthetic data. The fine-grained control offered by SoftPrompt, which provides the original soft label scores, uniquely enables such a targeted data curation approach. This strategy aims to remove potentially noisy or less informative examples generated from ambiguous soft label scores, thereby refining the synthetic dataset for improved downstream performance. The impact of varying the margin on model performance is



visualized in Figure 4.1 (see Appendix for full results in numbers, e.g., Tables A.1 and A.2).

Concretely, we define an ambiguity margin $m \in [0.0, 0.5)$ for binary tasks. Synthetic examples whose original soft label probability p_i (representing the score for the positive class) fell within the range $[0.5 - m, 0.5 + m]$ were excluded from the training set. We experimented with different values of $m \in \{0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3\}$.

As illustrated in Figure 4.1, applying this margin filter generally had a positive impact. For SUBJ, where unfiltered data ($m = 0.0$) showed declining performance with size, filtering was especially effective. Using SoftPrompt at $r = 100\%$, increasing the margin m to 0.2 improved accuracy to 83.9 from an unfiltered baseline of 61.2. This filtered performance significantly surpassed that of HardPrompt (59.5 at $r = 100\%$). With SoftPrompt+CoT on SUBJ at $r = 100\%$, the accuracy similarly increased from 69.9 (unfiltered) to 81.1 with an $m = 0.25$ filter. For IMDB, using SoftPrompt+CoT ($r = 100\%$), filtering with $m = 0.25$ increased accuracy from 89.3 to 91.8, nearing the Gold performance of 94.1. Similar positive trends with filtering were observed for SST, where SoftPrompt+CoT ($r = 100\%$) improved from 77.5 to 83.6 with an $m = 0.25$ filter. These results underscore the utility of filtering ambiguous examples when using SoftPrompt-generated data.

4.4 Leveraging soft labels directly in training

An inherent advantage of SoftPrompt is its generation of explicit soft label vectors $\mathbf{p}_i$, which provides flexibility for downstream model training. While



our initial experiments utilized binarized versions of these labels for direct comparison with baselines, the richer nuanced information encoded in $\mathbf{p}_i$ can be directly leveraged. To demonstrate the benefits of this approach, we experimented with training the logistic classifier by minimizing the Kullback-Leibler (KL) divergence between the model’s predicted probability distribution $\hat{\mathbf{p}}_i$ and the soft targets $\mathbf{p}_i$, on SUBJ and Emotion.

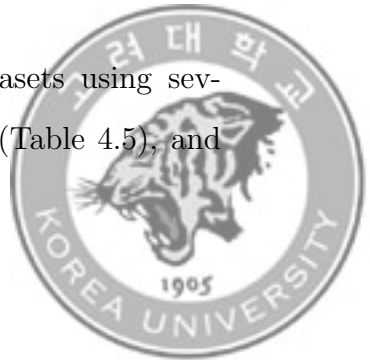
Table 4.3: Comparison of training with binarized and soft targets from SoftPrompt ($m = 0.0$).

Target	SUBJ		Emotion	
	Accuracy ($\uparrow$)	ECE ($\downarrow$)	Macro F1 ($\uparrow$)	ECE ($\downarrow$)
Binarized ($\text{argmax } \mathbf{p}_i$)	69.9 (± 5.9)	28.6 (± 0.1)	41.2 (± 0.4)	17.3 (± 0.4)
Soft ($\mathbf{p}_i$)	76.2 (± 3.1)	7.3 (± 0.5)	44.7 (± 0.3)	6.8 (± 0.2)

As shown in Table 4.3, training with soft targets yielded direct benefits over using binarized labels from the same SoftPrompt-generated data. For SUBJ, accuracy rose to 76.2 (from 69.9), while for Emotion, macro F1 increased to 44.7 (from 41.2). More significantly, model calibration improved dramatically: expected calibration error (ECE) on SUBJ dropped to 7.3 (from 28.6) and on Emotion to 6.8 (from 17.3). This demonstrates that SoftPrompt’s continuous labels enable more reliable probability estimates.

4.5 Diversity

We assessed the textual diversity of the generated datasets using several metrics: vocabulary size (Table 4.4), distinct 2-grams (Table 4.5), and



inter/intra-class pairwise cosine similarity (Table 4.6 and 4.7).

Across all datasets, the Gold (real) data exhibited the highest diversity, featuring the largest vocabulary sizes and highest distinct-2 n-gram ratios. Compared to HardPrompt, SoftPrompt generally produced text with richer vocabularies and higher distinct-2 n-gram counts, with SoftPrompt+CoT often yielding the most diverse synthetic text by these measures among the generation methods. For instance, on IMDB, SoftPrompt+CoT’s vocabulary size (13,843) was more than double that of HardPrompt (6,368) and significantly larger than HardPrompt+CoT (8,696). Similarly, its distinct-2 score (0.0273) surpassed HardPrompt (0.0090) and HardPrompt+CoT (0.0143).

In terms of embedding similarity, HardPrompt variants typically showed very high intra-class similarity, indicating less diversity within each class (e.g., for IMDB, SoftPrompt+CoT intra-class similarity was 0.7488 vs. 0.8619 for HardPrompt). SoftPrompt, particularly SoftPrompt+CoT, tended to reduce this intra-class similarity, suggesting more varied examples within classes, thereby moving closer to the characteristics of the Gold data, though all synthetic methods still showed higher similarity than Gold.

Table 4.4: Comparison of the *vocabulary size* of different datasets.

Method	IMDb	SST	SUBJ	Emotion	AGNews
Gold	77,169	18,267	20,189	17,094	102,166
HardPrompt	6,368	420	3,936	1529	4,595
HardPrompt+CoT	8,696	731	2,349	1788	6,377
SoftPrompt	11,283	1,123	4,075	8399	9,544
SoftPrompt+CoT	13,843	1,604	5,141	9,908	1,1862

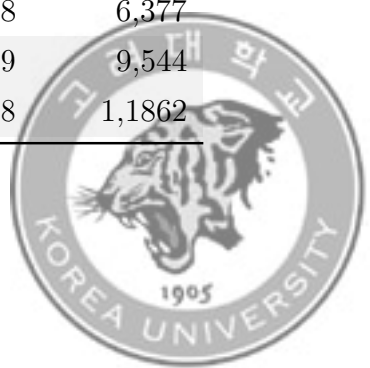


Table 4.5: Comparison of the distinct n -grams of different datasets ($n = 2$).

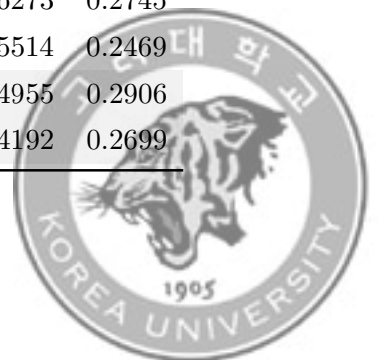
Method	IMDb	SST	SUBJ	Emotion	AGNews
Gold	0.1912	0.5625	0.5458	0.3486	0.2910
HardPrompt	0.0090	0.0064	0.0524	0.0054	0.0125
HardPrompt+CoT	0.0143	0.0187	0.0625	0.0072	0.0176
SoftPrompt	0.0150	0.0238	0.0822	0.0263	0.0336
SoftPrompt+CoT	0.0273	0.0447	0.1526	0.0208	0.0493

Table 4.6: Inter/intra-class pairwise similarity of binary tasks in the embedding space.

Method	IMDb			SST			SUBJ		
	Inter	Intra	Avg	Inter	Intra	Avg	Inter	Intra	Avg
Gold	0.3555	0.3950	0.3753	0.2531	0.2705	0.2534	0.1991	0.2520	0.2256
HardPrompt	0.4905	0.8619	0.6761	0.2897	0.7220	0.5058	0.0096	0.6827	0.3461
HardPrompt+CoT	0.5032	0.8315	0.6673	0.3494	0.7225	0.5359	0.0391	0.3481	0.1936
SoftPrompt	0.6766	0.8375	0.6768	0.4765	0.7201	0.4767	0.2000	0.4718	0.2003
SoftPrompt+CoT	0.6282	0.7488	0.6282	0.4841	0.6739	0.4843	0.1851	0.2811	0.1852

Table 4.7: Inter/intra-class and average pairwise similarity of multiclass tasks in the embedding space.

Method	Emotion			AGNews		
	Inter	Intra	Avg	Inter	Intra	Avg
Gold	0.2665	0.2874	0.2715	0.1587	0.2293	0.1764
HardPrompt	0.2931	0.6963	0.3603	0.1569	0.6273	0.2745
HardPrompt (CoT)	0.2724	0.6565	0.3364	0.1455	0.5514	0.2469
SoftPrompt	0.3248	0.5457	0.3612	0.2223	0.4955	0.2906
SoftPrompt+CoT	0.4297	0.4998	0.4414	0.2202	0.4192	0.2699



Chapter 5

Discussion

While SoftPrompt successfully improved synthetic data utility over hard-labeling, enhancing specific generative attributes, particularly diversity, remained a focus. Our exploration into this area through two distinct trials yielded unexpected outcomes, offering crucial insights into the complexities of fine-grained LLM control.

In preliminary experiments, we investigated *few-shot prompting*, hypothesizing that including varying numbers of real-world examples (1–50) in the prompt and instructing the LLM to “contrast and diversify” its outputs would guide it to generate more diverse and accurate synthetic texts, thereby boosting downstream classification performance compared to zero-shot generation. Surprisingly, empirical results showed no significant performance difference regardless of example count. This suggests that for complex generative tasks, examples might paradoxically act as constraints, inadvertently limiting the LLM’s natural diversity by encouraging replication of example-specific patterns rather than exploring the full semantic space dictated by the soft label condition. This finding aligns with observations from prior work such



as GPT3Mix [14], which reported that synthetic examples generated via few-shot prompting often inherit the syntactic and semantic structure of the input examples. The model tends to replicate local patterns and stylistic features, rather than explore broader variations—even in sentiment and phrasing—thus restricting generative diversity as it interpolates between familiar patterns instead of generalizing beyond them. As a direction for future work, we plan to conduct a direct quantitative benchmark against post-hoc, few-shot methods like GPT3Mix. This will allow us to empirically validate the practical advantages of our pre-hoc, zero-shot paradigm in terms of performance, diversity, and generation efficiency.

Second, to counter observed repetition in sequential generations, we explored *batch generation*, expecting that creating multiple texts per API call would reduce surface-level homogeneity. This method partially succeeded by decreasing repetitive phrasing. However, it introduced an unforeseen issue: texts generated in batches tended to be significantly shorter, regardless of how large a value we set for the max tokens. This uncontrolled length variation likely negated any diversity gains, resulting in no improvement in overall downstream performance. This trial underscores the difficulty of simultaneously controlling multiple interdependent generative attributes; optimizing one (repetition) can negatively impact another (length), highlighting the limitations of simple batching for complex control.

Collectively, these trials reveal that straightforward interventions like few-shot prompting or basic batching are insufficient for robust control over multifaceted generative qualities such as diversity and length. Successfully governing these attributes necessitates more sophisticated, perhaps multi-faceted



or iterative, control mechanisms integrated into the LLM generation process, guiding future research in this domain.

5.1 On the Utility of Training Specialized Classifiers vs. Direct LLM Classification

Given the powerful zero-shot classification abilities of modern LLMs, a valid and practical question is why one should go through the process of generating synthetic data to train a smaller, specialized classifier instead of using the LLM for direct classification. The goal of our framework is not to replace LLMs, but rather to leverage their powerful generative capabilities to build "specialist" models that are more suitable for real-world deployment. This approach offers four key advantages.

1. **Cost and Scalability:** Classifying millions of data points via LLM API calls is often prohibitively expensive. A trained specialist model, once deployed, can perform inference at a fraction of the cost, making it viable for large-scale applications.
2. **Inference Speed and Latency:** API-based classification can introduce significant network latency, which is unacceptable for real-time services. A locally deployed lightweight classifier can provide responses in milliseconds.
3. **Data Privacy and Security:** Using an external LLM API requires sending potentially sensitive data to a third party. An in-house model



trained on synthetic data keeps all information within the organization’s control.

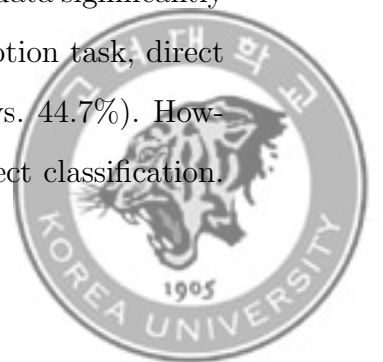
4. **Task-Specific Optimization:** While LLMs are excellent ”generalists,” they may not match the performance of a ”specialist” model fine-tuned for a specific, nuanced task, such as achieving better model **calibration** (ECE), as shown in Table 4.3.

To empirically investigate this trade-off, we conducted an additional experiment comparing the direct classification performance of our generator LLM (`gemini-2.0-flash`) against our method on the SUBJ and Emotion datasets. We used a simple zero-shot prompt instructing the LLM to classify the text and respond with only the label word. The results are summarized in Table 5.1.

Table 5.1: Comparison of direct LLM classification vs. training a specialist model on SoftPrompt-generated data.

Dataset	Method	Metric	Performance
SUBJ	LLM Direct Classification	Accuracy	70.0%
	Our Method (SoftPrompt+CoT, m=0.25)	Accuracy	81.1%
Emotion	LLM Direct Classification	Macro F1	50.4%
	Our Method (Soft Target Training)	Macro F1	44.7%

The results in Table 5.1 present a nuanced picture. For the SUBJ task, our method of training a specialist classifier on filtered synthetic data significantly outperformed direct classification by the LLM. For the Emotion task, direct LLM classification achieved a higher Macro F1 score (50.4% vs. 44.7%). However, we observed a crucial qualitative issue during this direct classification.



Despite being prompted with a clear task and a discrete set of categories, the LLM occasionally exhibited control instability by generating labels outside of the specified set (e.g., producing a non-existent category or a descriptive phrase instead of the required single-word label). While such errors could be mitigated with more sophisticated prompt engineering or output parsing, this process would incur additional costs and complexity at scale. In conclusion, the approach of using an LLM as a generator to train smaller, faster, safer, and cheaper specialist models remains a highly practical and often superior strategy for real-world applications. Our framework serves as a vital bridge between the immense intelligence of large models and the practical demands of industry.



Chapter 6

Conclusion

We introduced SoftPrompt, a novel framework conditioning large language models (LLMs) on soft labels—continuous-valued targets for more nuanced, expressive control than traditional hard labels. Encoding these soft labels directly in prompts, SoftPrompt generates realistic (text, label) pairs better reflecting real-world language’s ambiguity and gradience.

Across diverse classification tasks, SoftPrompt consistently outperformed hard-label baselines in generalization accuracy and text diversity, especially with data scarcity or subjectivity. Additionally, filtering ambiguous soft labels near decision boundaries significantly improved downstream performance, while soft label training (not binarized targets) enhanced model calibration. These capabilities position SoftPrompt as a strong candidate for controllable, data-efficient generation. Furthermore, soft label conditioning offers a promising path towards more realistic, effective synthetic NLP data.



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Appendix A

Margin-based filtering

Table A.1: Test accuracy on binary classification tasks with different margins (*without* CoT). The best value in each row is **boldfaced**, and the second best is underlined.

Data	Size	Margin (m)						
		0.00	0.05	0.10	0.15	0.20	0.25	0.30
IMDb	1%	75.1 (± 4.7)	76.6 (± 4.0)	76.5 (± 3.7)	80.6 (± 3.1)	86.9 (± 2.6)	<u>89.8</u> (± 2.3)	90.4 (± 0.8)
	5%	76.1 (± 4.7)	74.4 (± 3.9)	74.7 (± 4.6)	78.7 (± 2.9)	85.0 (± 3.6)	<u>89.8</u> (± 1.2)	90.4 (± 0.5)
	10%	76.2 (± 3.9)	74.5 (± 4.0)	75.5 (± 4.2)	78.0 (± 2.6)	84.8 (± 1.9)	<u>89.5</u> (± 0.7)	90.4 (± 0.5)
	25%	77.2 (± 2.9)	74.7 (± 3.2)	76.2 (± 2.5)	77.9 (± 2.2)	81.0 (± 3.5)	<u>87.8</u> (± 2.5)	90.1 (± 1.0)
	50%	77.1 (± 3.0)	74.8 (± 2.3)	76.1 (± 2.2)	78.0 (± 1.7)	85.7 (± 0.8)	<u>89.9</u> (± 0.4)	90.2 (± 0.2)
	100%	76.6 (± 1.9)	75.1 (± 1.7)	75.9 (± 1.4)	78.1 (± 0.8)	84.9 (± 0.6)	<u>89.3</u> (± 0.4)	90.3 (± 0.3)
SST	1%	68.9 (± 6.1)	68.9 (± 7.1)	70.1 (± 7.1)	72.3 (± 8.4)	<u>73.9</u> (± 9.6)	73.4 (± 9.6)	75.4 (± 9.9)
	5%	70.3 (± 4.1)	70.3 (± 3.5)	71.3 (± 5.3)	72.3 (± 5.1)	77.4 (± 7.3)	<u>77.4</u> (± 9.1)	77.1 (± 9.1)
	10%	71.1 (± 4.1)	70.1 (± 3.4)	71.5 (± 4.0)	73.7 (± 2.9)	77.5 (± 6.6)	79.1 (± 7.5)	<u>77.6</u> (± 7.7)
	25%	72.4 (± 3.9)	71.1 (± 3.4)	73.6 (± 2.5)	73.8 (± 1.9)	<u>78.4</u> (± 4.2)	81.4 (± 4.3)	78.0 (± 6.5)
	50%	73.7 (± 3.1)	72.2 (± 2.8)	75.3 (± 2.7)	74.5 (± 1.5)	78.9 (± 3.2)	<u>82.3</u> (± 3.7)	82.4 (± 3.3)
	100%	75.2 (± 2.5)	72.9 (± 2.2)	74.0 (± 1.9)	74.3 (± 0.9)	79.3 (± 2.2)	82.8 (± 2.6)	<u>81.8</u> (± 4.7)
SUBJ	1%	69.8 (± 8.3)	71.5 (± 7.2)	75.2 (± 7.8)	<u>78.4</u> (± 5.9)	78.5 (± 5.6)	75.5 (± 6.0)	76.2 (± 6.6)
	5%	67.1 (± 5.4)	67.0 (± 5.9)	70.7 (± 6.6)	79.2 (± 5.5)	83.4 (± 1.5)	<u>82.1</u> (± 2.1)	81.4 (± 1.4)
	10%	64.8 (± 4.0)	66.5 (± 4.2)	66.0 (± 3.7)	77.6 (± 5.7)	83.8 (± 1.3)	<u>83.0</u> (± 0.9)	82.1 (± 0.9)
	25%	62.5 (± 3.6)	65.1 (± 3.2)	64.8 (± 2.8)	74.0 (± 4.8)	83.9 (± 0.8)	<u>83.5</u> (± 0.6)	82.3 (± 0.6)
	50%	61.7 (± 3.0)	64.5 (± 2.2)	64.8 (± 2.4)	73.6 (± 2.7)	84.2 (± 0.5)	<u>83.5</u> (± 0.3)	82.3 (± 0.6)
	100%	61.2 (± 2.0)	64.0 (± 2.2)	63.4 (± 1.7)	70.2 (± 2.3)	83.9 (± 0.8)	<u>83.6</u> (± 0.3)	82.1 (± 0.5)

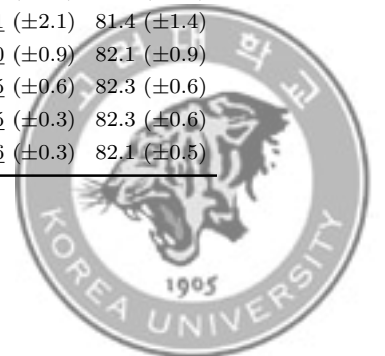


Table A.2: Test accuracy on binary classification tasks with different margins (*with* CoT). The best value in each row is **boldfaced**, and the second best is underlined.

Data	Size	Margin (m)						
		0.00	0.05	0.10	0.15	0.20	0.25	0.30
IMDb	1%	86.6 (± 2.3)	87.9 (± 2.2)	89.4 (± 1.3)	90.4 (± 1.0)	91.3 (± 0.3)	<u>91.3</u> (± 0.5)	90.2 (± 0.9)
	5%	86.8 (± 1.9)	87.6 (± 1.6)	88.9 (± 1.4)	90.4 (± 0.7)	91.6 (± 0.3)	<u>91.5</u> (± 0.2)	90.5 (± 0.5)
	10%	87.9 (± 1.5)	88.5 (± 1.1)	89.5 (± 0.0)	90.5 (± 0.6)	91.7 (± 0.2)	<u>91.6</u> (± 0.2)	90.6 (± 0.4)
	25%	88.9 (± 0.7)	88.5 (± 1.0)	89.2 (± 0.6)	90.1 (± 0.6)	91.7 (± 0.1)	<u>91.6</u> (± 0.1)	90.7 (± 0.6)
	50%	89.1 (± 0.7)	88.6 (± 0.8)	89.8 (± 0.4)	90.6 (± 0.3)	91.8 (± 0.1)	<u>91.7</u> (± 0.0)	90.5 (± 0.1)
	100%	89.3 (± 0.3)	89.0 (± 0.4)	89.6 (± 0.2)	90.5 (± 0.2)	91.8 (± 0.0)	<u>91.7</u> (± 0.0)	90.7 (± 0.1)
SST	1%	75.5 (± 3.2)	77.2 (± 2.8)	80.6 (± 2.1)	80.9 (± 2.1)	82.8 (± 1.5)	<u>83.7</u> (± 1.1)	84.0 (± 0.8)
	5%	75.2 (± 2.6)	77.3 (± 2.5)	80.4 (± 1.1)	81.2 (± 1.7)	82.8 (± 0.7)	<u>83.5</u> (± 0.6)	84.7 (± 0.3)
	10%	75.4 (± 2.9)	77.7 (± 2.2)	80.4 (± 1.0)	81.0 (± 0.7)	82.5 (± 0.6)	<u>83.3</u> (± 0.5)	84.7 (± 0.3)
	25%	76.4 (± 2.7)	76.9 (± 1.9)	80.4 (± 0.6)	81.1 (± 0.4)	82.6 (± 0.3)	<u>83.6</u> (± 0.3)	84.8 (± 0.2)
	50%	76.4 (± 2.1)	77.3 (± 1.3)	80.4 (± 0.4)	81.0 (± 0.2)	82.5 (± 0.3)	<u>83.5</u> (± 0.2)	84.8 (± 0.1)
	100%	77.5 (± 1.7)	77.5 (± 1.3)	80.3 (± 0.3)	80.9 (± 0.3)	82.7 (± 0.2)	<u>83.6</u> (± 0.2)	84.9 (± 0.1)
SUBJ	1%	71.5 (± 7.0)	73.5 (± 6.2)	76.3 (± 4.0)	76.1 (± 4.7)	77.6 (± 3.6)	77.2 (± 3.5)	77.2 (± 3.5)
	5%	76.0 (± 5.0)	77.6 (± 3.4)	78.8 (± 4.0)	79.4 (± 3.9)	80.5 (± 2.2)	<u>80.4</u> (± 1.8)	79.0 (± 1.7)
	10%	74.4 (± 5.1)	77.0 (± 4.1)	78.1 (± 3.2)	79.0 (± 2.9)	80.9 (± 1.5)	<u>80.8</u> (± 1.3)	78.9 (± 1.7)
	25%	74.0 (± 5.2)	75.4 (± 4.2)	77.3 (± 4.2)	77.6 (± 3.4)	<u>80.6</u> (± 1.3)	81.9 (± 1.0)	79.4 (± 1.0)
	50%	72.0 (± 6.3)	73.6 (± 4.4)	75.7 (± 3.5)	78.3 (± 2.6)	81.3 (± 0.9)	<u>81.2</u> (± 0.8)	79.2 (± 0.7)
	100%	69.9 (± 5.9)	72.3 (± 3.7)	74.8 (± 3.2)	77.6 (± 2.1)	80.7 (± 1.0)	81.1 (± 0.7)	78.8 (± 0.6)



Appendix B

Dataset Details

IMDb

The IMDb dataset is a large-scale sentiment classification corpus consisting of 50,000 highly polarized movie reviews, equally split into training and testing sets. It is commonly used to evaluate model performance on long-form text classification.

- **Source:** <https://huggingface.co/datasets/stanfordnlp/imdb>
- **Task:** Binary sentiment classification
- **Split:** 25,000 train / 25,000 test
- **Features:** review, label
- **Label:** 0 = negative, 1 = positive
- **Balance:** Equal class distribution
- **Preprocessing:** Neutral instances removed, reviews pre-cleaned



SST (Stanford Sentiment Treebank)

The SST dataset is a fine-grained sentiment analysis corpus that includes fully labeled parse trees, enabling detailed analysis of compositional sentiment in language. Each sentence is annotated with a continuous sentiment score ranging from 0.0 (very negative) to 1.0 (very positive). The dataset also includes sub-sentence (phrase) annotations and syntactic parse trees in both parent-pointer and Penn Treebank formats.

- **Source:** <https://huggingface.co/datasets/stanfordnlp/sst>
- **Task:** Sentiment scoring (regression) and classification
- **Split:** Train / Validation / Test for sentence-level data
- **Features:** sentence, label, tokens, tree, ptb_tree, phrase
- **Label:** Real-valued sentiment score $\in [0.0, 1.0]$
- **Configurations:**
 - *default*: sentence-level regression
 - *dictionary*: phrase-level annotations (no split)
 - *ptb*: full parse trees with discrete labels (0–4)

SUBJ (Subjectivity Classification)

The SUBJ dataset is a benchmark for classifying whether a sentence expresses a subjective opinion or an objective statement. It is widely used in tasks like sentence representation evaluation.



- **Source:** <https://huggingface.co/datasets/SetFit/subj>
- **Task:** Binary subjectivity classification
- **Split:** 8,000 train / 2,000 test
- **Features:** `text`, `label`, `label_text`
- **Label:** 0 = objective, 1 = subjective
- **Balance:** Balanced between classes

Emotion

The Emotion dataset contains short English tweets labeled with one of six basic emotions: *sadness*, *joy*, *love*, *anger*, *fear*, and *surprise*. It is used to evaluate fine-grained emotional understanding in social media texts.

- **Source:** <https://huggingface.co/datasets/dair-ai/emotion>
- **Task:** Multi-class emotion classification (6-way)
- **Split:** 16,000 train / 2,000 validation / 2,000 test
- **Unsplit:** 416,809 (14,972 examples for the minority *surprise* class)
- **Features:** `text`, `label`
- **Label Mapping:** 0 = sadness, 1 = joy, 2 = love, 3 = anger, 4 = fear, 5 = surprise
- **Balance:** Imbalanced



- **Note:** In our experiments, we used the *unsplit* set as gold training data, and the *split* set as the test set. Specifically, each class is downsampled to the minority class count.

AG News

The AG News dataset is a topic classification benchmark comprising news headlines and descriptions, categorized into four topics: *World*, *Sports*, *Business*, and *Science & Technology*. Only the title field is used in our experiments.

- **Source:** https://huggingface.co/datasets/SetFit/ag_news
- **Task:** Multi-class topic classification (4-way)
- **Split:** 120,000 train / 7,600 test (30,000 / 1,900 per class)
- **Features:** title, description, label
- **Label Mapping:** 0 = World, 1 = Sports, 2 = Business, 3 = Sci/Tech
- **Notes:** Description field not used in training. Fields enclosed in quotes; internal quotes escaped with `"`; line breaks encoded as `\n`



Appendix C

Prompts

```
1 prompt = """
2     You are tasked with generating realistic movie reviews
3     to train a sentiment classifier.
4     Use a sentiment scale from 0 (negative) to 1 (positive).
5     Reviews should generally be between 100 and 500 words,
6     avoiding overly short or excessively long responses.
7     Generate a movie review with a sentiment score of {score}.
8 """
```

Figure C.1: Prompt template used for IMDb

```
1 prompt = """
2     You are tasked with generating realistic movie review snippets
3     to train a sentiment classifier.
4     Use a sentiment scale from 0 (negative) to 1 (positive)
5     Each should generally be between 10 and 50 words,
6     usually a single sentence or a short phrase.
7     Use realistic language that one might find in a movie review.
8     Write a movie review with a sentiment score of {score}.
9 """
```

Figure C.2: Prompt template used for SST



```

1 prompt = """
2     You are simulating short social-media posts.
3     Use a subjectivity scale where 0 represents an objective statement
4     (e.g., factual descriptions, neutral narration) and 1 represents
5     a subjective statement (e.g., personal opinions, emotional expressions,
6     or evaluative language).
7     Use everyday language as if written on an online platform.
8     Follow the user instructions about which label to use.
9     Write a brief social-media post that corresponds to the score = {score}.
10  """

```

Figure C.3: Prompt template used for SUBJ

```

1 prompt = """
2     You are an AI language model assistant trained
3     to generate emotionally nuanced short messages.
4     You will receive an emotion distribution across six emotions
5     (sadness, joy, love, anger, fear, surprise),
6     where each value indicates the strength of that emotion and all sum to 1.
7     Your task is to generate a short, natural, Twitter-style message
8     that reflects this emotional distribution in a realistic and creative way.
9     Messages should generally be between 10 and 30 words.
10    Avoid generic or unnatural sentences.
11    Use human-like language with emotional subtlety.
12    Generate a short, natural Twitter-style message that reflects
13    the following emotional distribution: {score}
14  """

```

Figure C.4: Prompt template used for Emotion



```

1 prompt = """
2     You are an AI language model assistant trained
3     to generate realistic news headlines.
4     You will receive a topic distribution across four categories:
5     World, Sports, Business, and Science/Technology.
6     Each value represents the proportion of relevance for that topic,
7     and all values sum to 1.
8     Your task is to write a news headline that reflects this topic distribution.
9     Headlines should be written in a style typical of online news sources.
10    Headlines should generally be between 5 and 15 words,
11    but this can vary naturally based on the topic.
12    Do not mention the categories or percentages directly in the headline.
13    Avoid generic or unnatural language. Aim for specificity and realism.
14    Generate a realistic and coherent news headline
15    that reflects the following topic distribution: {score}
16    """

```

Figure C.5: Prompt template used for AG News



Appendix D

Qualitative Examples of SoftPrompt Outputs

To supplement the quantitative evaluation, this appendix presents qualitative examples of synthetic data generated by our **SoftPrompt+CoT** framework across all five benchmark datasets: IMDB, SST, SUBJ, Emotion, and AGNews. We focus exclusively on the SoftPrompt+CoT variant, which includes explicit Chain-of-Thought (CoT) reasoning before text generation. This allows readers to inspect not only the final output text $\mathbf{x}$, but also the model’s intermediate thought process—offering a transparent account of how the model interprets and translates soft label prompts into natural language. Each example comprises: (i) the soft label vector $\mathbf{p}$, (ii) the generated text $\mathbf{x}$, and (iii) the accompanying CoT reasoning. To support interpretability, we annotate key phrases in the generated text that correspond to the label distribution: **phrases aligned with the positive label** and **phrases indicative of the negative label**. These annotations help visualize how well the model integrates the intended label mixture into the generated content. Collectively, these ex-



amples provide qualitative evidence that soft label conditioning—when combined with explicit reasoning—enables fine-grained and interpretable control over text generation.



Table D.1: Representative **SoftPrompt+CoT**-generated IMDb reviews with different sentiment scores. Positive phrases are highlighted in blue, and negative phrases in red. Each example includes the model input (soft label), output (text), and rationale (reasoning).

Moderately Positive
Soft Label (p): positive: 0.77, negative: 0.23 Text (x): Overall, this movie was a delightful surprise. The acting was superb, particularly from the lead actress who truly embodied her character. The cinematography was also stunning, with several scenes that were visually breathtaking. The soundtrack perfectly complemented the mood of the film, enhancing the emotional impact. The plot was engaging and kept me guessing until the very end. However, the pacing felt a bit uneven at times. The second act dragged slightly, and some of the subplots could have been trimmed to maintain a more consistent flow. Additionally, while the ending was satisfying, it felt a little rushed. A few extra minutes to flesh out the resolution would have been beneficial. Despite these minor flaws, the movie is a thoroughly enjoyable experience. The strong performances, beautiful visuals, and captivating story outweigh the pacing issues. I would definitely recommend this film to anyone looking for a thought-provoking and entertaining movie. It's a solid 4 out of 5 stars! Reasoning: The review should express a generally positive sentiment, but not overwhelmingly so. Some minor flaws or areas for improvement should be mentioned to justify the 0.77 score. The movie's strengths should be highlighted first, followed by constructive criticism.
Predominantly Negative
Soft Label (p): positive: 0.27, negative: 0.73 Text (x): I walked into the theater with high hopes for this movie, given the hype and the star-studded cast. Unfortunately, what I experienced was a colossal disappointment. The plot was convoluted and nonsensical, jumping from one scene to another without any clear direction. I found myself constantly struggling to understand what was happening, and by the end, I was just completely lost. The acting was subpar, with most of the actors seeming disinterested and phoning in their performances. There were a few exceptions, but they were not enough to save the film. And the special effects, which I had expected to be groundbreaking, were surprisingly cheap and unconvincing. Overall, this movie was a complete waste of time and money. I regret spending my evening watching it, and I would not recommend it to anyone. Save yourself the disappointment and skip this one. Reasoning: The overall sentiment is negative (0.27 is closer to 0 than to 1). Start with a premise of high expectations that were not met. Mention specific aspects of the movie that were disappointing, such as the plot, acting, and special effects. Use negative language to describe these aspects. Conclude with a statement that expresses overall disappointment and regret for watching the movie.



Table D.2: Representative **SoftPrompt+CoT**-generated SST snippets with varying sentiment scores. Positive phrases are highlighted in **blue**, and negative phrases in **red**. Each example includes the model input (soft label), output (text), and rationale (reasoning).

Moderately Negative Soft Label (p): positive: 0.31, negative: 0.69 Text (x): The plot was meandering and confusing, but the cinematography was beautiful. Overall, a disappointing experience with a few bright spots. I can't recommend it strongly, but it wasn't a total waste of time either. Reasoning: The sentiment is slightly negative. The review should reflect disappointment and some redeeming qualities, but overall leaning towards negative.
Slightly Positive Soft Label (p): positive: 0.57, negative: 0.43 Text (x): It's not a masterpiece, but it has some redeeming qualities. The plot is a bit convoluted, but the acting is decent enough to keep you engaged. Overall, a passable film for a rainy afternoon. Reasoning: The sentiment is slightly positive. The review should acknowledge both good and bad aspects of the movie, but ultimately lean towards a slightly favorable impression. It should be short and concise.
Clearly Positive Soft Label (p): positive: 0.79, negative: 0.21 Text (x): A surprisingly good movie with great acting and an engaging story. It has some flaws, but overall a worthwhile watch! Reasoning: The movie was surprisingly good, the acting was great, and the story was engaging. It has some flaws, but overall it's a good movie.



Table D.3: Representative **SoftPrompt+CoT**-generated SUBJ examples with varying degrees of subjectivity. Subjective phrases are highlighted in blue, and objective phrases in red. Each example includes the model input (soft label), output (text), and rationale (reasoning).

Highly Objective
Soft Label (p): subjective: 0.03, objective: 0.97 Text (x): A pen is a writing instrument used to apply ink to a surface, usually paper, for writing or drawing. Reasoning: The goal is to create a factual statement with minimal subjective content. I will describe a common object and one of its basic functions. I will avoid evaluative language or personal opinions.
Mixed Subjectivity
Soft Label (p): subjective: 0.57, objective: 0.43 Text (x): The bus arrived 10 minutes late today. It wasn't a huge deal, but it did make me slightly anxious about being late for my appointment. Reasoning: The text should contain a mix of factual information and personal opinion. I will describe a common everyday scenario, and include a mild personal sentiment.
Clearly Subjective
Soft Label (p): subjective: 0.71, objective: 0.29 Text (x): That new coffee shop downtown has a surprisingly cozy atmosphere, though their latte art could use some work. I'd recommend it for a chill afternoon read! Reasoning: The text should express a personal feeling or opinion but not be overly emotional or aggressive. It should use descriptive language that suggests a viewpoint without being explicitly biased.

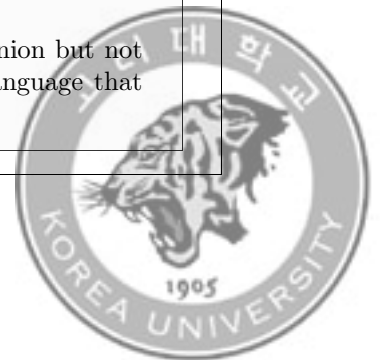


Table D.4: Representative **SoftPrompt+CoT**-generated examples from the Emotion dataset. Each phrase is annotated with its dominant emotional tone using six distinct colors: **sadness**, **joy**, **love**, **anger**, **fear**, and **surprise**.

Joy with Fear and Surprise
Soft Label (p): sadness: 0.03, joy: 0.47, love: 0.03, anger: 0.05, fear: 0.30, surprise: 0.13 Text (x): I could barely contain my excitement as I approached the old wooden gate. Today was the day! Butterflies fluttered in my stomach, a joyful dance anticipating the adventure ahead. As I pushed open the gate, a gust of wind slammed it shut behind me with a loud BANG! My heart leaped into my throat, a cold dread washing over me. For a moment, I stood frozen, fear gripping me. Then, a figure emerged from behind a tree – my best friend, grinning, holding a bouquet of my favorite flowers! The surprise quickly dissolved the fear, replaced by a wave of relief and even greater joy. What a thrilling, slightly terrifying, but ultimately wonderful way to start the day! Reasoning: The text should be predominantly joyful with a significant element of fear, a touch of surprise, and minimal sadness, anger, and love. I will craft a narrative that initially builds excitement and anticipation, then introduces an element of unexpected fear or suspense, before resolving into a positive, albeit slightly altered, joyful outcome. The surprise will be linked to the twist causing the fear.
Joy and Love
Soft Label (p): sadness: 0.03, joy: 0.50, love: 0.41, anger: 0.03, fear: 0.03, surprise: 0.02 Text (x): My heart is overflowing with happiness and love for you. Just thinking about you brings so much joy to my life, it's a truly wonderful feeling! Reasoning: The emotion distribution indicates a high degree of joy and love, with very little sadness, anger, fear, and surprise. The text should reflect these dominant positive emotions.



Table D.5: Representative **SoftPrompt+CoT**-generated examples from the AGNews dataset. Each phrase is annotated with the most relevant topic using: **World**, **Sports**, **Business**, and **Science/Technology**.

Primarily Business with Sports Angle Soft Label (p): world: 0.00, sports: 0.23, business: 0.77, sci/tech: 0.00 Text (x): Nike Shares Surge as Sneaker Sales Drive Unexpected Earnings Beat; Analysts Cite Resurgent Demand and Innovative Marketing Campaigns Targeting Gen Z. Reasoning: The topic distribution is heavily skewed towards business, with a smaller sports component. A headline should reflect a business story that might have a sports angle. I will generate a headline about a sports apparel company exceeding expectations.
Primarily World with Business Secondary Soft Label (p): world: 0.86, sports: 0.00, business: 0.14, sci/tech: 0.00 Text (x): Tensions Rise as Diplomatic Talks Stall Between World Leaders; Global Markets Brace for Uncertainty. Reasoning: The topic distribution indicates a strong focus on world news (85.9%), a moderate emphasis on business (14%), and negligible influence from sports and science/technology. I will craft a headline that primarily reflects a significant global event with potential business implications.
World with Sci/Tech Element Soft Label (p): world: 0.70, sports: 0.00, business: 0.00, sci/tech: 0.30 Text (x): International Tensions Rise as New AI Surveillance System Deployed at Border Crossings, Sparking Privacy Concerns and Diplomatic Protests. Reasoning: The topic distribution indicates a strong emphasis on world news (70.4%) and a smaller focus on science/technology (29.6%). Therefore, the headline should reflect a global event with a technological aspect or implication.

